

MySQL Concepts with Queries and Examples

This document covers core MySQL concepts with explanations, syntax, sample queries, and examples.

1. Create Database

Explanation:

This section demonstrates 1. create database with a practical example.

```
CREATE DATABASE company_db;  
USE company_db;
```

2. Create Table

Explanation:

This section demonstrates 2. create table with a practical example.

```
CREATE TABLE employees (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100),  
    email VARCHAR(100) UNIQUE,  
    gender ENUM('Male','Female','Other'),  
    salary DECIMAL(10,2),  
    department VARCHAR(50),  
    joining_date DATE  
);
```

3. Insert Data

Explanation:

This section demonstrates 3. insert data with a practical example.

```
INSERT INTO  
employees(name,email,gender,salary,department,joining_date)  
VALUES ('John','john@example.com','Male',50000,'IT','2024-01-10'),  
('Alice','alice@example.com','Female',65000,'HR','2023-08-15'),  
('David','david@example.com','Male',70000,'Finance','2022-12-20');
```

4. SELECT

Explanation:

This section demonstrates 4. select with a practical example.

```
SELECT * FROM employees;  
SELECT name,salary FROM employees;
```

5. WHERE

Explanation:

This section demonstrates 5. where with a practical example.

```
SELECT * FROM employees WHERE salary>60000;
```

6. ORDER BY

Explanation:

This section demonstrates 6. order by with a practical example.

```
SELECT * FROM employees ORDER BY salary DESC;
```

7. LIMIT

Explanation:

This section demonstrates 7. limit with a practical example.

```
SELECT * FROM employees LIMIT 5;
```

8. DISTINCT

Explanation:

This section demonstrates 8. distinct with a practical example.

```
SELECT DISTINCT department FROM employees;
```

9. UPDATE

Explanation:

This section demonstrates 9. update with a practical example.

UPDATE employees SET salary=75000 WHERE id=1;

10. DELETE

Explanation:

This section demonstrates 10. delete with a practical example.

DELETE FROM employees WHERE id=3;

11. ALTER TABLE

Explanation:

This section demonstrates 11. alter table with a practical example.

ALTER TABLE employees ADD phone VARCHAR(15);

ALTER TABLE employees DROP COLUMN phone;

12. DROP/TRUNCATE

Explanation:

This section demonstrates 12. drop/truncate with a practical example.

TRUNCATE TABLE employees;

DROP TABLE employees;

13. Aggregate Functions

Explanation:

This section demonstrates 13. aggregate functions with a practical example.

SELECT COUNT(*)

total,SUM(salary),AVG(salary),MAX(salary),MIN(salary) FROM
employees;

14. GROUP BY

Explanation:

This section demonstrates 14. group by with a practical example.

SELECT department,COUNT(*) employees FROM employees GROUP BY department;

15. HAVING

Explanation:

This section demonstrates 15. having with a practical example.

SELECT department,AVG(salary) avg_salary FROM employees GROUP BY department HAVING AVG(salary)>60000;

16. LIKE

Explanation:

This section demonstrates 16. like with a practical example.

SELECT * FROM employees WHERE name LIKE 'A%';

17. BETWEEN

Explanation:

This section demonstrates 17. between with a practical example.

SELECT * FROM employees WHERE salary BETWEEN 50000 AND 70000;

18. IN

Explanation:

This section demonstrates 18. in with a practical example.

SELECT * FROM employees WHERE department IN ('IT','HR');

19. JOIN Example

Explanation:

This section demonstrates 19. join example with a practical example.

```
CREATE TABLE departments(  
id INT PRIMARY KEY,  
department_name VARCHAR(50)
```

);

```
SELECT e.name,d.department_name  
FROM employees e  
INNER JOIN departments d  
ON e.department=d.department_name;
```

20. LEFT JOIN

Explanation:

This section demonstrates 20. left join with a practical example.

```
SELECT e.name,d.department_name FROM employees e LEFT JOIN  
departments d ON e.department=d.department_name;
```

21. RIGHT JOIN

Explanation:

This section demonstrates 21. right join with a practical example.

```
SELECT e.name,d.department_name FROM employees e RIGHT JOIN  
departments d ON e.department=d.department_name;
```

22. SELF JOIN

Explanation:

This section demonstrates 22. self join with a practical example.

```
SELECT a.name,b.name FROM employees a JOIN employees b ON  
a.id<>b.id;
```

23. UNION

Explanation:

This section demonstrates 23. union with a practical example.

```
SELECT name FROM employees UNION SELECT department_name  
FROM departments;
```

24. Views

Explanation:

This section demonstrates 24. views with a practical example.

```
CREATE VIEW employee_view AS SELECT name,salary FROM  
employees;  
SELECT * FROM employee_view;
```

25. Index

Explanation:

This section demonstrates 25. index with a practical example.

```
CREATE INDEX idx_name ON employees(name);
```

26. Constraints

Explanation:

Constraints enforce data integrity. PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, DEFAULT, CHECK, AUTO_INCREMENT

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27. Foreign Key

Explanation:

This section demonstrates 27. foreign key with a practical example.

```
CREATE TABLE orders(  
id INT AUTO_INCREMENT PRIMARY KEY,  
employee_id INT,  
FOREIGN KEY(employee_id) REFERENCES employees(id)  
);
```

28. Transactions

Explanation:

This section demonstrates 28. transactions with a practical example.

```
START TRANSACTION;  
UPDATE employees SET salary=salary+1000 WHERE id=1;  
COMMIT;  
-- or ROLLBACK;
```

29. Subquery

Explanation:

This section demonstrates 29. subquery with a practical example.

```
SELECT * FROM employees WHERE salary>(SELECT AVG(salary) FROM  
employees);
```

30. CASE

Explanation:

This section demonstrates 30. case with a practical example.

```
SELECT name, CASE WHEN salary>60000 THEN 'High' ELSE 'Normal'  
END AS salary_level FROM employees;
```

31. Date Functions

Explanation:

This section demonstrates 31. date functions with a practical example.

```
SELECT CURDATE(),NOW(),YEAR(joining_date),MONTH(joining_date) FROM  
employees;
```

32. String Functions

Explanation:

This section demonstrates 32. string functions with a practical example.

```
SELECT UPPER(name),LOWER(name),LENGTH(name),CONCAT(name,'  
- ',department) FROM employees;
```

33. Numeric Functions

Explanation:

This section demonstrates 33. numeric functions with a practical example.

```
SELECT ROUND(123.456,2),CEIL(10.2),FLOOR(10.9),ABS(-50);
```

36. Practice Queries

Explanation:

This section demonstrates 36. interview practice queries with a practical example.

-- Second Highest Salary

```
SELECT MAX(salary) FROM employees  
WHERE salary<(SELECT MAX(salary) FROM employees);
```

-- Duplicate Emails

```
SELECT email,COUNT(*) FROM employees  
GROUP BY email  
HAVING COUNT(*)>1;
```

-- Top 3 Highest Salaries

```
SELECT * FROM employees ORDER BY salary DESC LIMIT 3;
```

Practice Exercises

1. Create Student and Course tables.
2. Apply Primary Key and Foreign Key.
3. Insert 10 records.
4. Write JOIN queries.
5. Find highest, second highest salary.
6. Use GROUP BY and HAVING.